

4. The longitudinal motion of an aerial vehicle is given by

$$\begin{aligned}\dot{w} &= -2w + 197\dot{\theta} - 27\delta \\ \ddot{\theta} &= -0.25w - 15\dot{\theta} - 45\delta\end{aligned}$$

- (a) Pick an appropriate set of states and express the dynamics of the system in state-space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}.$$

- (b) Determine the stability of the open-loop system.

- (c) Determine the proportional state feedback control law

$$\mathbf{u} = \mathbf{K}\mathbf{x}$$

required such that the closed-loop system response is oscillatory with a damping ratio $\zeta = 0.5$ and an undamped natural frequency $\omega_n = 20$ rad/s.

(a) $\mathbf{x} = [w \ \dot{\theta}]$ and $\mathbf{u} = [\delta]$

$\therefore \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$

$\therefore \begin{bmatrix} \dot{w} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} -2 & 197 \\ -0.25 & -15 \end{bmatrix} \begin{bmatrix} w \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} -27 \\ -45 \end{bmatrix} \delta$

(b) For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$

At equilibrium state, $\dot{\mathbf{x}}_0 = \mathbf{A}\mathbf{x}_0 = \mathbf{0}$

$\therefore \begin{bmatrix} \dot{w}_0 \\ \ddot{\theta}_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} w_0 \\ \dot{\theta}_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Adding small perturbation,

$\dot{\mathbf{x}}_0 + \dot{\mathbf{x}}' = \mathbf{A}(\mathbf{x}_0 + \mathbf{x}') = \mathbf{x}'\mathbf{A}_{\mathbf{x}_0}$

$\therefore \begin{bmatrix} \dot{w}' \\ \ddot{\theta}' \end{bmatrix} = \begin{bmatrix} -2 & 197 \\ -0.25 & -15 \end{bmatrix} \begin{bmatrix} w' \\ \dot{\theta}' \end{bmatrix}$

Let $x' = x^* e^{\lambda t}$ where x^* is constant

$$\therefore \dot{x}' = \lambda x^* e^{\lambda t} = \lambda x'$$

$$\therefore (A_{x_0} - \lambda I) x' = 0$$

$$\therefore |A_{x_0} - \lambda I| = 0$$

$$\therefore \begin{vmatrix} -2 - \lambda & 197 \\ -0.25 & -15 - \lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 + 17\lambda + 30 + \frac{197}{4} = 0$$

$$\therefore \lambda = -\frac{17}{2} \pm \frac{\sqrt{17^2 - 317}}{2} = -\frac{17}{2} \pm \sqrt{7}i$$

$\therefore$ oscillatory stable

$$\text{for } \lambda_1 = -\frac{17}{2} + \sqrt{7}i, \xi_1 = \begin{bmatrix} 1 \\ -\frac{13}{394} + \frac{1}{197}\sqrt{7}i \end{bmatrix}$$

$$\text{for } \lambda_2 = -\frac{17}{2} - \sqrt{7}i, \xi_2 = \begin{bmatrix} 1 \\ -\frac{13}{394} - \frac{1}{197}\sqrt{7}i \end{bmatrix}$$

$$\therefore x(t) = k_1 \xi_1 e^{\lambda_1 t} + k_2 \xi_2 e^{\lambda_2 t}$$

$$= k_1 \begin{bmatrix} 1 \\ -\frac{13}{394} + \frac{\sqrt{7}}{197}i \end{bmatrix} e^{(-\frac{17}{2} + \sqrt{7}i)t} + k_2 \begin{bmatrix} 1 \\ -\frac{13}{394} - \frac{\sqrt{7}}{197}i \end{bmatrix} e^{(-\frac{17}{2} - \sqrt{7}i)t}$$

$$\therefore t_{\pm} = \frac{\ln 2}{-Re(\lambda)} = 0.08155 \text{ s}$$

$$\omega_d = \sqrt{7} \text{ rad/s}$$

$$\omega_n = \frac{\sqrt{317}}{2} \text{ rad/s}$$

$$\therefore T = \frac{2\pi}{\omega_n} = 0.7058 \text{ s}$$

$$\zeta = \frac{-Re(\lambda)}{\omega_n} = 0.9548$$

$$(c) u = K_p x = Kx = [K_w \ K_{\dot{\theta}}] x$$

$$\therefore \dot{x} = Ax + B(r - y) = Bx + Br - BKx$$

$\therefore$ Repeat process above,

$$(A - BK - \lambda I) x' = 0$$

$$\therefore |A - BK - \lambda I| = 0$$

$$\therefore \begin{vmatrix} -2 - 27K_w - \lambda & 197 - 27K_{\dot{\theta}} \\ -0.25 - 45K_w & -15 - 45K_{\dot{\theta}} - \lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 + (-27K_w - 45K_{\dot{\theta}} - 17)\lambda$$

$$+ (1215K_{\dot{\theta}}K_w + 405K_w + 90K_{\dot{\theta}} + 30)$$

$$- (1215K_{\dot{\theta}}K_w + \frac{27}{4}K_w - 8865K_{\dot{\theta}} - \frac{197}{4}) = 0$$

$$\therefore \lambda^2 + (27K_w + 45K_\theta + 17)\lambda + \left(\frac{1647}{4}K_w - 8715K_\theta - \frac{77}{4}\right) = 0$$

$$\therefore \lambda = -\frac{27K_w + 45K_\theta + 17}{2}$$

$$\pm \frac{\sqrt{(-27K_w - 45K_\theta - 17)^2 - (1647K_w - 8715K_\theta - 77)}}{2}$$

$$= \lambda \pm si \quad \text{where } \lambda < 0$$

$$\therefore \lambda^2 + s^2 = 20^2 \text{ rad}^2/\text{s}^2 = 400 \text{ rad}^2/\text{s}^2$$

$$\therefore \zeta = \frac{-\lambda}{\omega_n} = 0.5 \quad \therefore -\lambda = 10 \text{ rad/s}$$

$$\therefore 27K_w + 45K_\theta + 17 = 20$$

$$\therefore \lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2$$

$$\therefore \frac{1647}{4}K_w - 8715K_\theta - \frac{77}{4} = \omega_n^2 = 400 \text{ rad}^2/\text{s}^2$$

$$\therefore K_w = 0.0342 \text{ and } K_\theta = 0.0462$$

5. Plot the state space structure for the system

$$\dot{x} = y$$

$$\dot{y} = \cos x$$

$$\dot{\mathbf{x}} = \mathbf{R}(\mathbf{x})$$

$$\text{At equilibrium, } \dot{\mathbf{x}}_0 = \mathbf{R}(\mathbf{x}_0) = \mathbf{0}$$

$$\therefore [\dot{x}_0 \ \dot{y}_0]^T = \mathbf{0} \quad \text{and} \quad [y_0 \ \cos x_0]^T = \mathbf{0}$$

$$\therefore y_0 = 0, \quad x_0 = \pm \frac{1}{2}n\pi$$

$$\text{Adding small perturbation, } \dot{\mathbf{x}}' = \mathbf{X}' \mathbf{R}_{\mathbf{x}_0}$$

$$\text{for } \mathbf{x}_0 = [\frac{1}{2}n\pi \ 0]^T,$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{for } \mathbf{x}_0 = [-\frac{1}{2}n\pi \ 0]^T,$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{Let } \mathbf{x}' = \mathbf{x}^* e^{\lambda t} \quad \text{where } \mathbf{x}^* \text{ is constant}$$

$$\therefore \dot{\mathbf{x}}' = \lambda \mathbf{x}^* e^{\lambda t} = \lambda \mathbf{x}' \quad \therefore (\mathbf{R}_{\mathbf{x}_0} - \lambda \mathbf{I}) \mathbf{x}' = \mathbf{0}$$

$$\therefore |R_{x_0} - \lambda I| = 0$$

$$\text{for } x_0 = [\tfrac{1}{2}n\pi \ 0]^T,$$

$$\begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = 0 \quad \therefore \lambda^2 + 1 \quad \therefore \lambda = \pm i$$

$$\therefore \xi_1 = \begin{bmatrix} 1 \\ i \end{bmatrix} \quad \text{and} \quad \xi_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

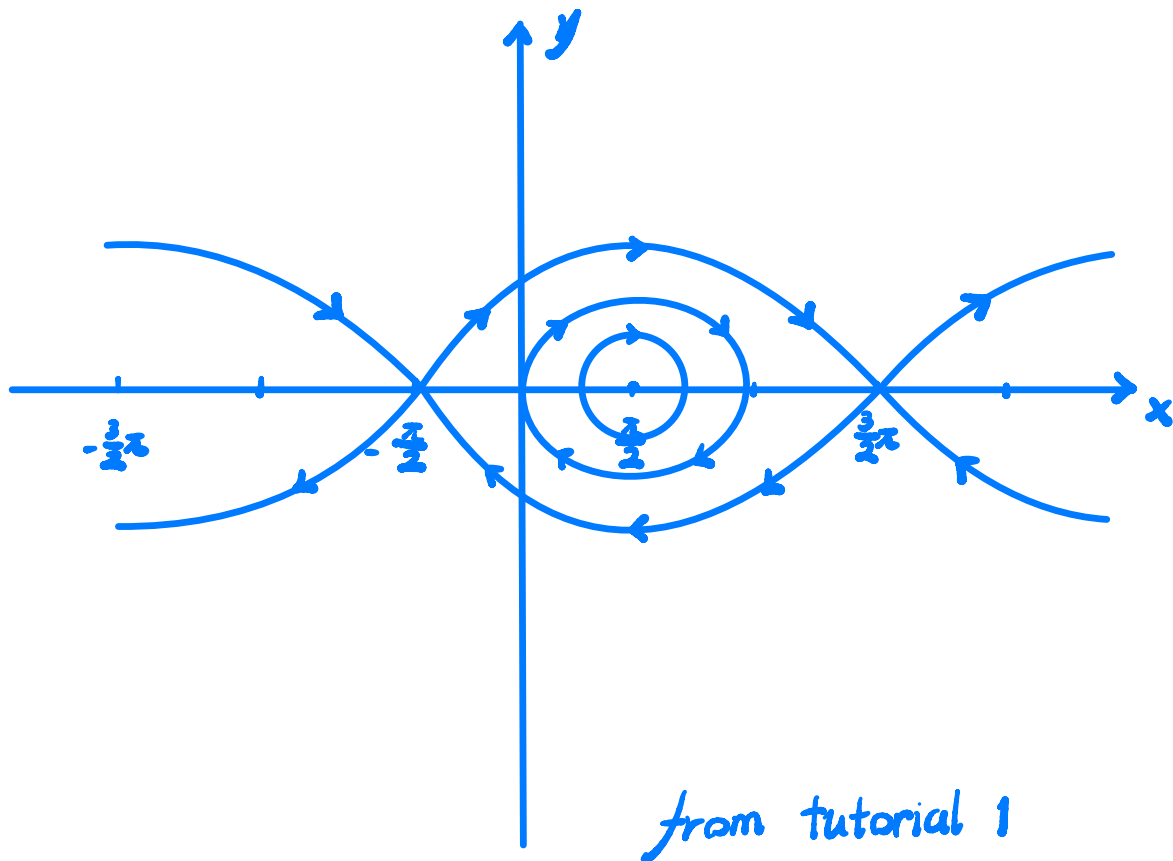
$$\therefore x_1(t) = k_{11} \begin{bmatrix} 1 \\ i \end{bmatrix} e^{it} + k_{12} \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{-it}$$

$$\text{for } x_0 = [-\tfrac{1}{2}n\pi \ 0]^T,$$

$$\begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \quad \therefore \lambda^2 - 1 \quad \therefore \lambda = \pm 1$$

$$\therefore \xi_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \xi_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore x_2(t) = k_{21} \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + k_{22} \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$



Why it is clockwise for neutral oscillation?

$$\dot{x} = y$$

$$\dot{y} = \cos x$$

$$\text{Let } y = 0.000001$$

$$\text{therefore } \dot{x} = 0.000001$$

adding small perturbation Δt .

$$\text{so } \Delta x = \dot{x} \Delta t = 0.000001 \Delta t$$